

VLIELAND, Netherlands

WMO: 062420

Lat: 53.2400N

Lon: 4.9208E

Elev: 5

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-4.7	-2.9	-8.2	1.9	-3.6	-6.2	2.2	-2.3	20.1	7.1	18.8	7.0	7.5	90	0.800

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.4	24.9	19.9	22.8	18.9	21.3	17.9	20.5	23.9	19.4	22.1	18.5	20.8	5.3	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	13.9	22.6	18.2	13.1	21.4	17.4	12.4	20.3	59.0	24.1	55.3	22.2	52.4	20.8	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	16.1	14.4	DB	-6.2	29.0	2.5	1.5	-8.1	30.1	-9.6	31.0	-11.0	31.9	-12.8	33.0	(4)
(5)			WB	-6.8	22.6	2.5	0.8	-8.6	23.1	-10.1	23.6	-11.5	24.0	-13.3	24.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	3.8	3.9	5.8	9.0	12.3	15.0	17.4	17.9	15.7	11.9	8.1	5.5		
	DBStd	5.70	3.57	3.21	2.69	2.84	2.78	2.29	2.35	2.15	2.18	2.61	2.85	3.32		
	HDD10.0	778	192	172	131	51	8	0	0	0	0	13	69	142		
	HDD18.3	2886	450	406	388	279	187	104	48	34	84	199	308	399		
	CDD10.0	979	0	0	1	22	80	150	228	244	170	72	11	1		
	CDD18.3	46	0	0	0	0	1	4	18	19	4	0	0	0		
	CDH23.3	139	0	0	0	0	4	16	62	52	6	0	0	0		
	CDH26.7	18	0	0	0	0	0	2	10	6	1	0	0	0		
(14) Wind	WSAvg	7.5	8.6	8.0	7.6	6.8	7.1	6.8	6.5	6.5	7.1	8.0	8.1	8.7		
(15) Precipitation	PrecAvg	820	68	54	44	39	47	56	70	85	87	96	93	81		
	PrecMax	1016	127	121	91	83	84	96	158	189	174	230	173	152		
	PrecMin	574	9	19	5	1	18	21	21	17	11	30	12	34		
	PrecStd	121	32	24	24	20	18	22	35	48	44	47	35	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	10.4	13.7	20.1	23.2	26.0	27.9	27.3	23.8	18.6	14.8	12.0		
		MCWB	9.5	8.6	10.4	14.1	16.8	19.9	21.7	21.1	19.4	16.2	13.5	10.8		
	2%	DB	9.6	9.2	11.2	16.7	20.3	22.1	24.9	24.6	21.3	17.2	13.7	11.1		
		MCWB	8.5	8.0	9.0	12.4	15.5	18.1	19.8	20.2	18.0	15.3	12.4	9.9		
	5%	DB	8.8	8.3	9.9	14.8	18.2	20.2	22.6	22.6	20.0	16.2	12.7	10.2		
		MCWB	7.7	7.3	8.3	11.6	14.8	17.0	18.8	19.1	17.2	14.5	11.3	9.1		
	10%	DB	8.0	7.6	9.0	13.0	16.4	18.6	21.0	21.1	18.9	15.4	11.8	9.3		
		MCWB	7.1	6.6	7.7	10.6	13.6	15.8	17.9	18.1	16.4	13.8	10.2	8.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	9.0	11.0	14.6	17.8	20.5	22.4	21.8	19.9	16.7	13.8	11.2		
		MCDB	10.5	9.8	13.0	18.7	21.1	24.5	27.3	25.6	22.9	18.3	14.5	11.7		
	2%	WB	8.8	8.2	9.4	13.0	16.2	18.7	20.3	20.7	18.5	15.6	12.8	10.2		
		MCDB	9.4	9.0	10.7	16.2	19.4	21.4	23.7	23.9	20.7	16.9	13.6	10.9		
	5%	WB	7.9	7.5	8.6	11.7	15.0	17.2	19.2	19.6	17.6	14.9	11.5	9.3		
		MCDB	8.6	8.3	9.6	14.3	17.9	19.7	21.9	22.2	19.5	15.9	12.5	10.0		
	10%	WB	7.2	6.8	7.8	10.7	13.8	16.1	18.2	18.6	16.8	14.1	10.6	8.4		
		MCDB	7.9	7.5	8.8	12.8	16.1	18.2	20.6	20.7	18.5	15.2	11.6	9.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.2	3.5	4.3	5.5	5.4	5.5	5.3	5.4	4.9	4.2	3.4	3.3		
		MCDBR	2.8	3.5	5.7	8.5	8.4	8.4	8.3	7.9	6.3	4.6	3.4	3.1		
	5% WB	MCWBR	3.1	3.0	4.1	5.1	4.9	5.2	4.6	4.5	4.1	3.8	3.3	3.3		
		MCWBR	2.9	3.4	5.2	7.7	7.8	7.6	7.4	7.3	5.7	4.5	3.5	3.0		
(40) Clear-Sky Solar Irradiance	taub		0.333	0.350	0.377	0.404	0.403	0.401	0.411	0.403	0.387	0.365	0.343	0.324		
	taud		2.317	2.347	2.300	2.274	2.346	2.365	2.346	2.379	2.415	2.411	2.351	2.310		
	Ebn at Noon		629	745	800	826	847	851	834	819	785	719	612	562		
	Edh at Noon		61	80	102	119	116	115	116	106	91	75	59	52		
(44) All-Sky Solar Radiation	RadAvg		0.61	1.25	2.60	4.28	5.32	5.54	5.31	4.43	3.06	1.68	0.71	0.45		
	RadStd		0.06	0.21	0.24	0.40	0.32	0.35	0.53	0.33	0.26	0.15	0.06	0.05		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page